Regression Modelling and Classification

Group Assignment – Group 1

Note: Part A and Part B are two separate assignments (Assignment 3 – Regression, Assignment 4 – Classification). Each should be developed and submitted as its own repository – see Section 6.

1 Objective

A retail company wants to optimize its advertising expenditure and improve loan risk assessment using predictive analytics.

2 Dataset Description

Part A – Regression (Assignment 3)

Dataset: Advertising Dataset (ISLR) – `https://www.statlearning.com/resources-first-edition`
Direct file: `https://www.statlearning.com/s/Advertising.csv`
Kaggle mirror: `https://www.kaggle.com/datasets/ashydv/advertising-dataset`
Variables include:

- TV Advertising Budget
- Radio Advertising Budget
- Newspaper Advertising Budget
- Product Sales

Part B – Classification (Assignment 4)

Dataset: Give Me Some Credit (Kaggle) – `https://www.kaggle.com/c/GiveMeSomeCredit`
Target Variable: Loan Default (Yes/No)
Predictors include age, income, debt ratio, credit utilization, and number of open credit lines.

3 Analytical Tasks

Part A – Regression

1. Perform exploratory data analysis.
2. Build Simple and Multiple Linear Regression models.
3. Evaluate using: R^2, Adjusted R^2, RMSE, MAE.
4. Check regression assumptions: Linearity, Normality, Homoscedasticity, Multicollinearity (VIF).
5. Identify the most influential advertising channels.
6. Recommend an advertising budget strategy.

Part B – Classification

1. Prepare the dataset.
2. Develop: Logistic Regression, Decision Tree.
3. Evaluate using: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix.
4. Compare thresholds (0.30, 0.50, 0.70).
5. Identify important predictors.
6. Recommend a loan approval threshold.

4 Discussion

Answer the following:

1. Which advertising channel has the greatest impact on sales?
2. Which regression model performs better?
3. Which classification model performs better?
4. Which threshold would you recommend for loan approval?

5. How could AI improve this analysis?

5 Deliverables

1. Python Notebook
2. Tables and Visualizations
3. Technical Report (4–6 pages)
4. Executive Presentation (8 slides maximum)

6 Repository Structure

Since Part A and Part B are separate assignments, maintain two independent repositories rather than one combined repo:

- **Assignment 3 – Regression**: notebook, EDA plots, regression diagnostics, report, and slides for the Advertising dataset.
- **Assignment 4 – Classification**: notebook, EDA plots, model comparison, report, and slides for the Give Me Some Credit dataset.

Each repo should have its own `README.md` (objective, dataset link, how to run), `requirements.txt`, and a `report/` and `slides/` folder, so the two assignments can be graded and versioned independently.